Introduction to Computer Architecture

Sheet #7

Exercises:

1. Assume you have a machine that uses 32-bit integers and you are storing the hex value 1234 at address 0.
   a. Show how this is stored on a big endian machine.
   b. Show how this is stored on a little endian machine.
   c. If you wanted to increase the hex value to 123456, which byte assignment would be more efficient, big or little endian? Explain your answer.

2. Show how the following values would be stored by machines with 32-bit words, using little endian and then big endian format. Assume each value starts at address 10_{16}. Draw a diagram of memory for each, placing the appropriate values in the correct (and labeled) memory locations.
   a. $456789A1_{16}$
   b. $0000058A_{16}$
   c. 14148888_{16}

3. The first two bytes of a 2M x 16 main memory have the following hex values:
   - Byte 0 is FE
   - Byte 1 is 01

   If these bytes hold a 16-bit two's complement integer, what is its actual decimal value if:

   a. Memory is big endian?
   b. Memory is little endian?

4. Convert the following expressions from infix to reverse Polish (postfix) notation.
   a. X * Y + W * Z + V * U
   b. W * X + W * (U * V + Z)
   c. (W * (X + Y * (U * V)))/(U * (X + Y))

5. Convert the following expressions from reverse Polish notation to infix notation.
   a. W X Y Z - + *
   b. U V W X Y Z + * + * +
   c. X Y Z + V W - * Z + +

6. a. Write the following expression in postfix (Reverse Polish) notation. Remember the rules of precedence for arithmetic operators!

$$X = \frac{A - B + C * (D * E - F)}{G + H * K}$$

b. Write a program to evaluate the above arithmetic statement using a stack organized computer with zero-address instructions (so only pop and push can access memory).